# Building Vibration Measurement Analysis

Laboratory ranking, source signatures, and optical-table performance

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|---|---|
| **Source workbook:** | `FinalReportData.xlsx` |
| **Measurements:** | 15 spectra in 5 measured rooms |
| **Spectral range:** | 0–250.25 Hz at 0.48877 Hz resolution |
| **Report date:** | July 23, 2026 |

Prepared as a data-driven assessment for vibration-sensitive experimental work

# 1 Evidence for the heating plant

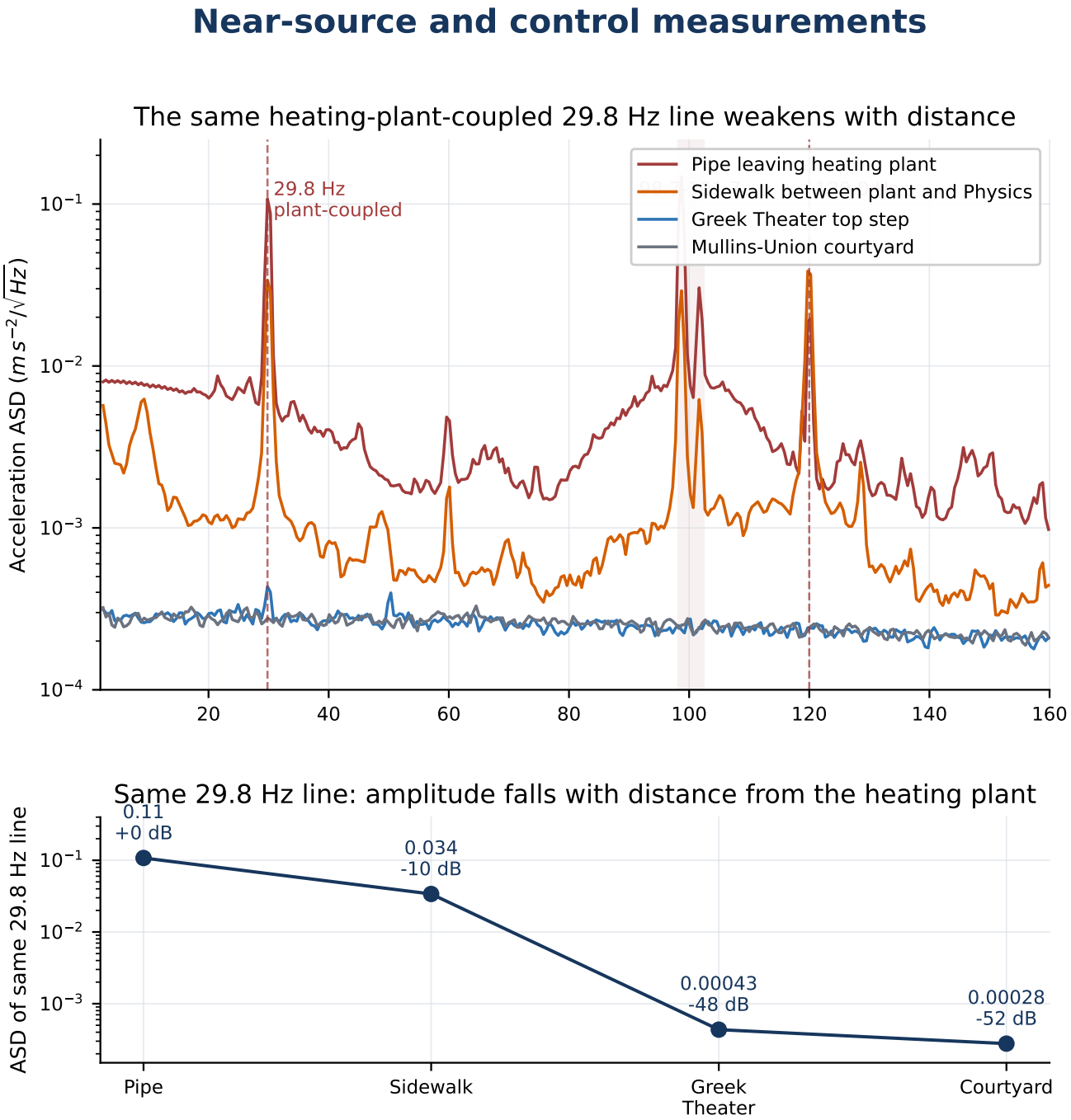



**Figure 1:** The same heating-plant-coupled 29.8 Hz line falls from $0.108\,\mathrm{m\,s^{-2}\,Hz^{-0.5}}$ on the plant pipe to $0.0339\,\mathrm{m\,s^{-2}\,Hz^{-0.5}}$ on the sidewalk, then approaches the remote background. The same near-to-far pattern is visible at other plant-associated frequencies.

The common frequency and its approximately 52.0 dB drop from the pipe to the courtyard identify the 29.8 Hz peak seen inside Physics as **the same vibration coupled from the heating plant**, not separate room-generated peaks or a sensor resonance.

## 2 Heating-plant spectral fingerprint

**Table 1:** Acceleration ASD at recurring plant-associated frequencies. All entries are in $10^{-3}$ $\mathrm{m\,s^{-2}\,Hz^{-0.5}}$. Values are read at the common frequency bin nearest each listed line.

| Frequency | Plant pipe | Sidewalk | Herzog | PHYS 111 | Titan | PHYS 245 | PHYS 244 |
|---|---|---|---|---|---|---|---|
| 29.8 Hz | 107.68 | 33.90 | 22.87 | 17.97 | 5.15 | 2.19 | 1.41 |
| 98.7 Hz | 148.13 | 29.14 | 3.09 | 1.65 | 0.26 | 0.28 | 0.23 |
| 101.7 Hz | 30.43 | 6.21 | 0.74 | 0.46 | 0.21 | 0.29 | 0.32 |
| 120.0 Hz | 18.86 | 38.58 | 2.41 | 0.59 | 0.23 | 0.26 | 0.58 |

The 29.8 Hz row tracks the same heating-plant-coupled vibration from the plant pipe into the Physics rooms. Its matching frequency and spatial decay are the principal source evidence. The group near 98.7 Hz to 102 Hz and the 120 Hz line provide additional corroboration. Exact amplitudes need not scale monotonically at every frequency because propagation, local structural resonances, and operating conditions differ among measurements.

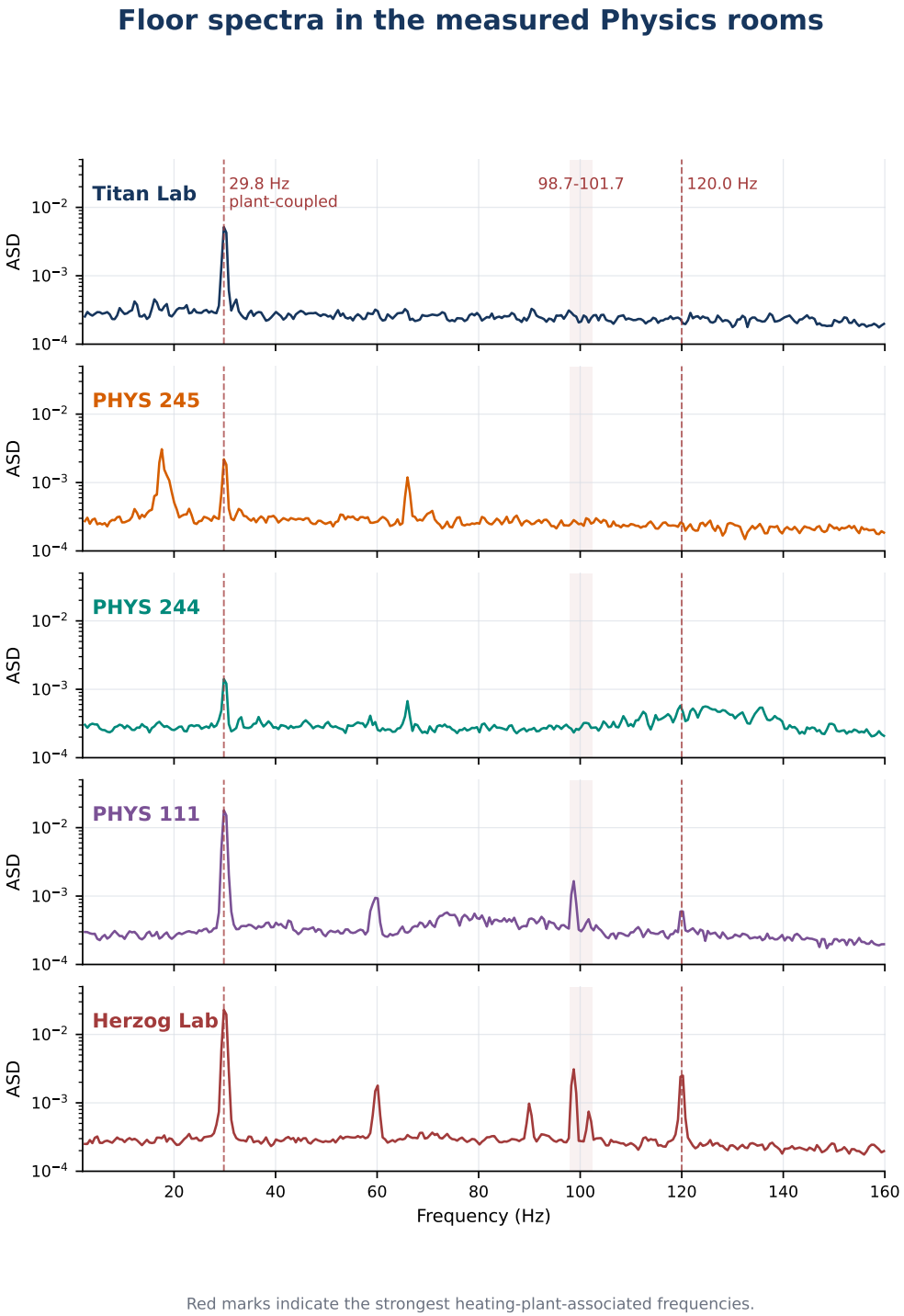



**Figure 2:** Floor spectra for the five measured Physics rooms, kept on common axes. The red marker at 29.8 Hz follows the same vibration coupled from the heating plant; the other red markers identify additional plant-associated frequencies.

# 3 Room choice

### Titan has the lowest background, but not the smallest plant-coupled 29.8 Hz line

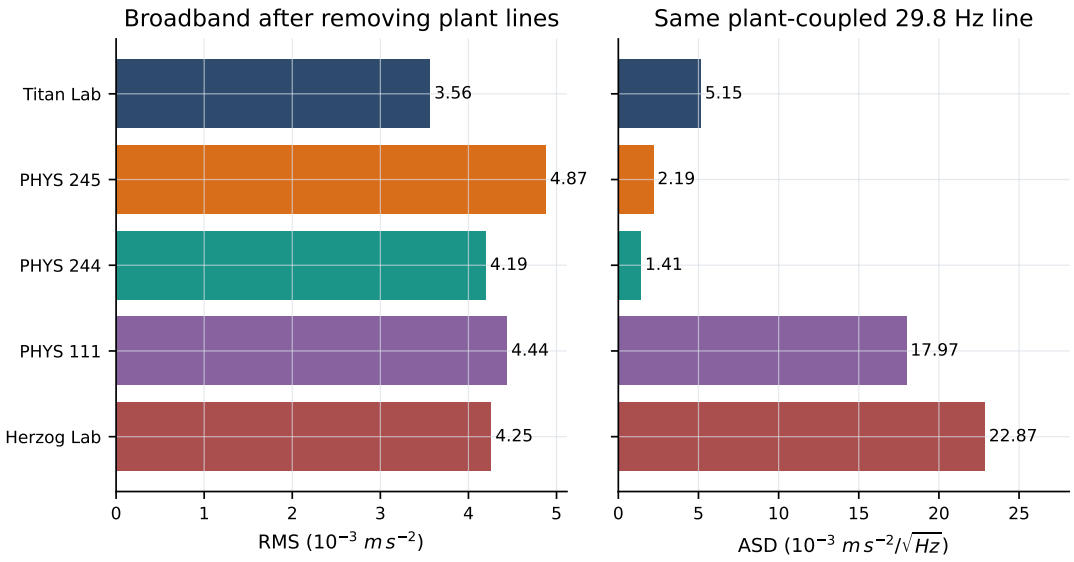



**Figure 3:** Two complementary room metrics. The left panel removes narrow plant-associated lines to compare broadband background. The right panel compares the amplitude of the same heating-plant-coupled 29.8 Hz line in each room.

The left panel explains why Titan appears so quiet in the spectra: after the coherent plant lines are set aside, it has the lowest band-integrated background of the five rooms. The right panel prevents that result from being overstated. PHYS 244 and PHYS 245 receive the same 29.8 Hz vibration at smaller amplitude, while its amplitude is substantially larger in Herzog and PHYS 111.

**Where I would build an optics laboratory:** I would consider Titan. It does have a significantly stronger 30Htz line from the plant, and that needs to be considered as that low frequency likes to go right through floated legs, but a floated table will help dampen it out somewhat and I suspect an ACTIVELY damped table will have little issues making it all but disappear–albeit at some expense. The rest of the spectrum is exceedingly low in the Titan lab so it was very well isolated for that TEM that was there some time ago. PHYS 245 would be my next choice and could be preferable for apparatus mechanically sensitive to this same 29.8 Hz vibration or if active dampening is not an option at which point you basically want as far away from the heating plant as possible and 245 gets you there.

# 4 What the optical tables accomplish

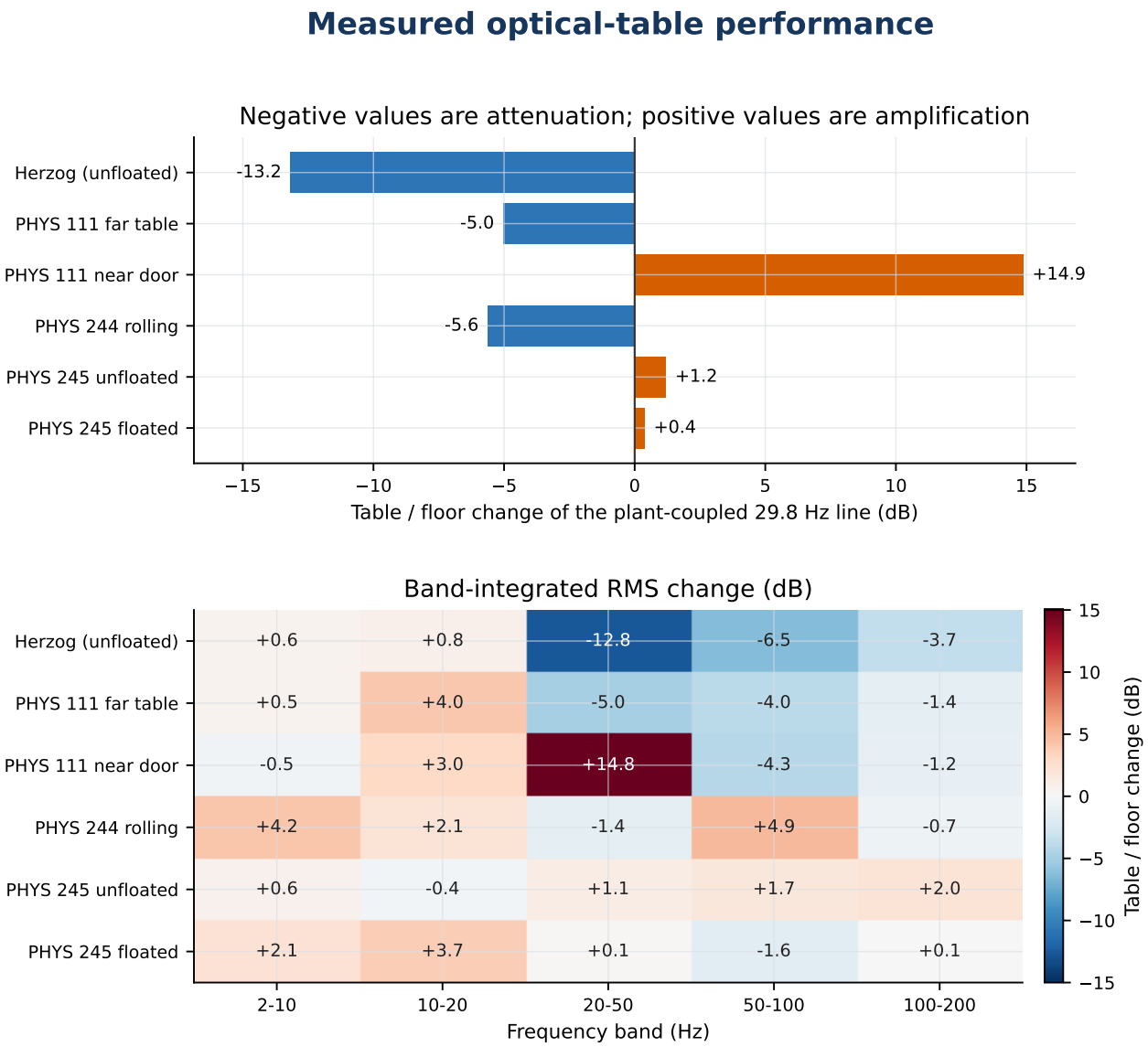



**Figure 4:** Measured table-to-floor change. Negative values mean attenuation; positive values mean amplification. The upper panel isolates the same heating-plant-coupled 29.8 Hz line; the lower panel summarizes broader bands.

The plot shows the measured change from each room floor to its table. Herzog has the largest reduction of the heating-plant-coupled 29.8 Hz line, which is not surprising because this line is much larger on the Herzog floor to begin with. Its remaining table amplitude is about $0.005\,00\,\mathrm{m\,s^{-2}\,Hz^{-0.5}}$: lower than the PHYS 111 table measurements, but about twice the PHYS 245 table level and roughly seven times the PHYS 244 rolling-table level at this frequency.

## 5 Herzog: reduced, but still exposed

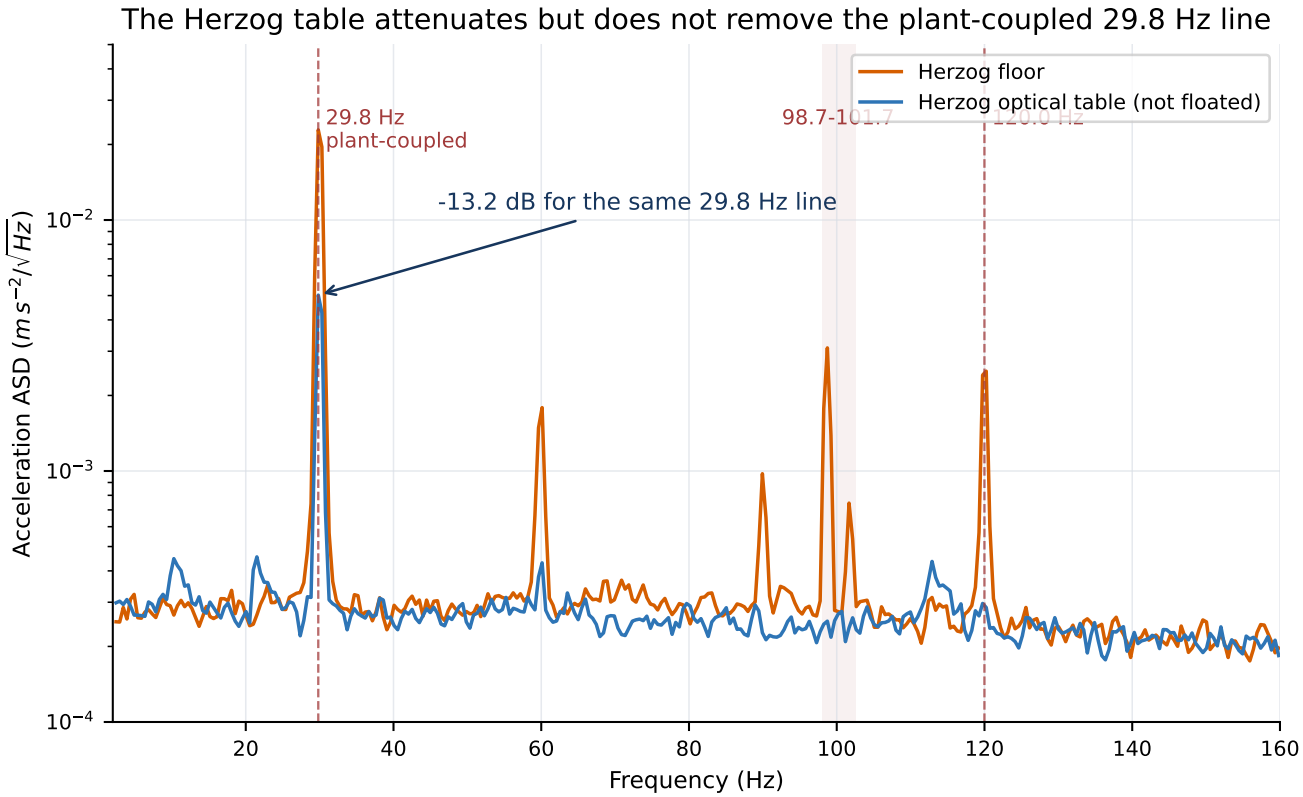



**Figure 5:** Herzog floor and optical-table spectra only. The freezer-top and back-counter traces are intentionally omitted.

The plot shows that the unfloated Herzog table reduces the same heating-plant-coupled 29.8 Hz floor line by 13.2 dB, but leaves a clear residual amplitude of approximately $0.005\,00\,\mathrm{m\,s^{-2}\,Hz^{-0.5}}$.

## Concise conclusion

The spatial controls, the plant-pipe measurement, and the recurring spectral fingerprint together make the heating plant the almost-certain source region for the low-frequency vibration observed in Herzog. The turbine is the most plausible individual source given the timing of the problem, but a short generator-free measurement tied to turbine operating state would provide the cleanest final confirmation. For room selection, Titan has the best broadband background; PHYS 245 receives the same heating-plant-coupled 29.8 Hz vibration at smaller amplitude and is the principal alternative as that frequency is especially difficult for an optical table to isolate. Obviously there is way in Hades I would put an optics lab in the Herzog lab. The very unfortunate thing for the entire building is that 29.8Hz is evidently difficult to filter even with a floated table.